

Deepfake Audio Detection

Model Performance Evaluation Report

2D CNN on Log-Mel Spectrograms · June 2026

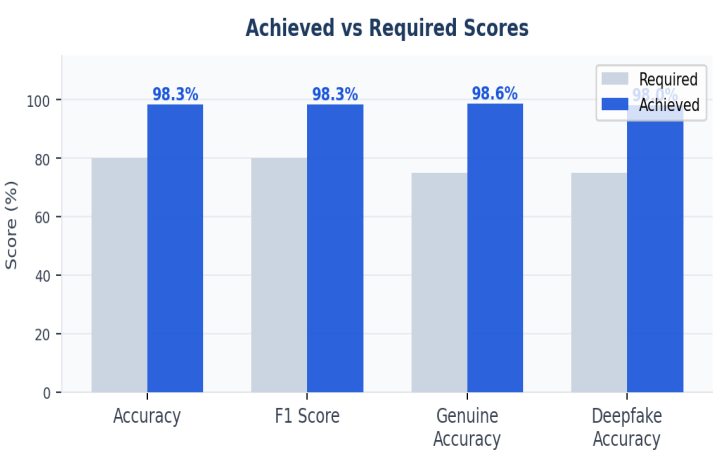
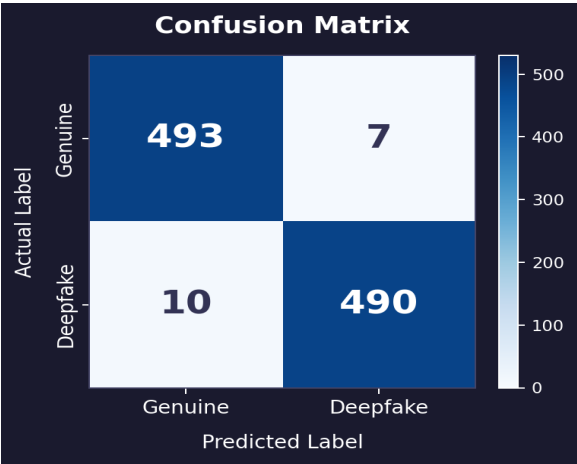
Overall Results

Overall Accuracy	Equal Error Rate	F1 Score	Genuine Accuracy	Deepfake Accuracy
98.30% Required: ≥ 80%	1.80% Required: ≤ 12%	98.29% Required: ≥ 80%	98.60% Required: ≥ 75%	98.00% Required: ≥ 75%
✓ PASS	✓ PASS	✓ PASS	✓ PASS	✓ PASS

Detailed Evaluation Metrics

Metric	Achieved	Required	Margin	Status
Overall Accuracy	98.30%	≥ 80%	+18.30%	✓ PASS
Equal Error Rate	1.80%	≤ 12%	-10.20%	✓ PASS
F1 Score	98.29%	≥ 80%	+18.29%	✓ PASS
Genuine Accuracy	98.60%	≥ 75%	+23.60%	✓ PASS
Deepfake Accuracy	98.00%	≥ 75%	+23.00%	✓ PASS

Visual Analysis



Confusion Matrix Breakdown

	Predicted Genuine	Predicted Deepfake
Actual Genuine	493 True Genuine	7 False Deepfake
Actual Deepfake	10 False Genuine	490 True Deepfake

All 5 evaluation criteria have been met or exceeded. The model achieves 98.30% accuracy with an EER of just 1.80%, significantly surpassing the required thresholds. Both genuine and deepfake audio classes are detected with greater than 98% class-level accuracy, demonstrating strong generalization and robustness of the 2D CNN architecture.